

TYLER FRANKE

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EDUCATION

Universität Hamburg

Dec. 2024 - Dec. 2027 (expected)

Doctor of Philosophy

Project A4 within CRC 1624 "Higher structures, moduli spaces and integrability"

Thesis title: Integrable systems and defect skein theory

Supervised by Prof. Dr. Ingo Runkel

University of Melbourne

Mar. 2021 - Dec. 2023

Master of Science (Mathematics and Statistics) (with Distinction)

Mathematical Physics and Physical Combinatorics specialisation

Thesis title: String order in the q -deformed Affleck-Kennedy-Lieb-Tasaki model

Supervised by Dr. Thomas Quella

University of Melbourne

Jun. 2021 - Dec. 2023

Diploma in Languages

German Studies specialisation

University of Melbourne

Mar. 2017 - Dec. 2019

Bachelor of Science

Mathematical Physics major

PUBLICATIONS AND PREPRINTS

- T. Franke, T. Quella, *Duality and hidden symmetry breaking in the q -deformed Affleck-Kennedy-Lieb-Tasaki model*, doi: 10.21468/SciPostPhysCore.7.4.078.

SELECTED PRESENTATIONS

- *Turaev-Viro TQFT's and the relation with the WRT TQFT*, BTW6: The Volume Conjecture: From Some TQFTs to Recent Developments (Fréjus, France), May 2026.
- *Constructions of braided module categories*, Algebra and mathematical physics research seminar, Universität Hamburg (Hamburg, Germany), May 2026.
- *The reflection equation*, Junior ZMP Seminar, Deutsches Elektronen-Synchrotron (Hamburg, Germany), January 2026.
- *Integrable systems and skein theory*, The Physics and Mathematics of Boundaries, Impurities, and Defects, gong-show presentation, Isaac Newton Institute (Cambridge, United Kingdom), September 2025.
- *Dualities in the q -deformed Affleck-Kennedy-Lieb-Tasaki model*, Tensor Categories, Quantum Symmetries, and Mathematical Physics, contributed talk, Matrix Institute (Creswick, Australia), November 2024.

TEACHING

- *Einführung in die Kategorientheorie*, bachelor seminar coordinator, Universität Hamburg, Winter Semester '26/'27.
- *Methods of mathematical physics*, teaching assistant, University of Melbourne, '23 and '24.
- *Linear algebra*, teaching assistant, University of Melbourne, '24.
- *Calculus 2*, teaching assistant, University of Melbourne, '23.